A complemented basic-subsequence subcase of the weak-null tensor question

FA/Banach run packet for arXiv:2305.06089

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Source question

The source paper is J. Rodriguez and A. Rueda Zoca, *Weak precompactness in projective tensor products*, arXiv:2305.06089. In Question `Gonzalo`, the authors ask:

> Let X and Y be Banach spaces. Let (x_n) and (y_n) be weakly null sequences in X and Y, respectively, such that $(x_n \otimes y_n)$ is weakly convergent in $X\widehat{\otimes}_\pi Y$. Is $(x_n \otimes y_n)$ weakly null in $X\widehat{\otimes}_\pi Y$?

Let us obtain another consequence of Theorem 3.8 in the context of weakly null sequences in projective tensor products. Observe that the usual basis of ℓ_2 shows that, in general, if $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ are weakly null sequences in X and Y, respectively, the sequence $(x_n \otimes y_n)_{n\in\mathbb{N}}$ may fail to be weakly null in $X\widehat{\otimes}_\pi Y$. To the best of our knowledge, the following question is open in complete generality.

Question 3.9. *Let X and Y be Banach spaces. Let $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ be weakly null sequences in X and Y, respectively, such that $(x_n \otimes y_n)_{n\in\mathbb{N}}$ is weakly convergent in $X\widehat{\otimes}_\pi Y$. Is $(x_n \otimes y_n)_{n\in\mathbb{N}}$ weakly null in $X\widehat{\otimes}_\pi Y$?*

Theorem 3.8 allows us to provide the following partial affirmative answer.

Corollary 3.10. *Let X and Y be Banach spaces such that X has property (R_p) and Y has property (R_q) for some $1 < p, q < \infty$ satisfying $1/p + 1/q \geq 1$. Let $(x_n)_{n\in\mathbb{N}}$*

Idea of the proof

The earlier approximation-property subcase kills the weak limit once the ambient projective tensor product embeds into the corresponding injective tensor product. The point here is that the whole ambient factor need not have the approximation property. If, after passing to a subsequence, one factor sequence spans a complemented subspace E with the approximation property, then the tensor subsequence is trapped in the complemented tensor subspace $E\widehat{\otimes}_\pi Y$. Inside $E\widehat{\otimes}_\pi Y$, the approximation-property argument forces the weak limit to be zero. The projection onto the complemented tensor subspace then forces the original ambient weak limit to be zero as well.

The theorem

Theorem 1. *Let X and Y be Banach spaces. Let $(x_n) \subset X$ and $(y_n) \subset Y$ be weakly null sequences. Suppose that $(x_n \otimes y_n)$ is weakly convergent in $X\widehat{\otimes}_\pi Y$.*

Assume that one of the following two conditions holds:

(a) some subsequence of (x_n) has closed linear span E, where E is complemented in X and has the approximation property;

(b) some subsequence of (y_n) has closed linear span F, where F is complemented in Y and has the approximation property.

Then $(x_n \otimes y_n)$ is weakly null in $X\widehat{\otimes}_\pi Y$.

In particular, it is enough in either condition to assume that the corresponding subsequence is basic and has complemented closed span.

Proof. Let $u \in X\widehat{\otimes}_\pi Y$ be the weak limit of $x_n \otimes y_n$. It is enough to prove $u = 0$.

We prove the case (a); the proof of (b) is symmetric. Pass to the subsequence in (a), and relabel it as (x_{n_k}). Let $E = \overline{\operatorname{span}}\{x_{n_k} : k \in \mathbb{N}\}$, let $i : E \to X$ be the inclusion, and choose a bounded projection $P : X \to E$. Since $Px_{n_k} = x_{n_k}$, the bounded projection

$$(iP) \otimes \mathrm{id}_Y : X\widehat{\otimes}_\pi Y \longrightarrow X\widehat{\otimes}_\pi Y$$

fixes every tensor $x_{n_k} \otimes y_{n_k}$. By weak continuity it fixes their weak limit:

$$((iP) \otimes \mathrm{id}_Y)u = u. \tag{1}$$

Now apply the bounded operator $P \otimes \mathrm{id}_Y : X\widehat{\otimes}_\pi Y \to E\widehat{\otimes}_\pi Y$. The subsequence $x_{n_k} \otimes y_{n_k}$ converges weakly in $E\widehat{\otimes}_\pi Y$ to

$$v := (P \otimes \mathrm{id}_Y)u.$$

The sequence (x_{n_k}) is weakly null as a sequence in E: if $e^* \in E^*$, then $e^* \circ P \in X^*$, so $e^*(x_{n_k}) = (e^* \circ P)(x_{n_k}) \to 0$. Also (y_{n_k}) remains weakly null in Y.

Because E has the approximation property, the canonical map $E\widehat{\otimes}_\pi Y \to E\widehat{\otimes}_\varepsilon Y$ is injective. For every $e^* \in E^*$ and $y^* \in Y^*$,

$$(e^* \otimes y^*)(v) = \lim_k e^*(x_{n_k})y^*(y_{n_k}) = 0.$$

Thus the image of v in $E\widehat{\otimes}_\varepsilon Y$ has injective norm zero. Injectivity gives $v = 0$.

Finally,

$$u = ((iP) \otimes \mathrm{id}_Y)u = (i \otimes \mathrm{id}_Y)(P \otimes \mathrm{id}_Y)u = (i \otimes \mathrm{id}_Y)v = 0$$

by (1). Hence the original weak limit is zero, so $(x_n \otimes y_n)$ is weakly null.

If the subsequence in (a) is basic, its closed span has a Schauder basis and therefore has the approximation property. The same applies in case (b). $\square$

Corollary 1. *Suppose that X has the following property: every seminormalized weakly null sequence in X admits a complemented basic subsequence. Then Question `Gonzalo` has an affirmative answer for X paired with any Banach space Y. The symmetric statement holds if Y has this property.*

Proof. Let $(x_n) \subset X$ and $(y_n) \subset Y$ be weakly null, and suppose that $x_n \otimes y_n$ is weakly convergent in $X\widehat{\otimes}_\pi Y$, with weak limit u. If (x_n) has a norm-null subsequence, then the corresponding tensor subsequence is norm-null, since weakly null sequences are bounded. Therefore $u = 0$.

Otherwise, (x_n) has a seminormalized subsequence. By the stated property of X, that subsequence has a complemented basic subsequence. Theorem 1 gives $u = 0$. The proof for Y is identical. $\square$

Verification and limitations

The proof uses only three standard facts: bounded operators induce bounded operators on projective tensor products; a complemented subspace gives a bounded tensor projection; and a Banach space with a Schauder basis has the approximation property. The final injectivity step is the usual approximation-property characterization that the canonical map $E\widehat{\otimes}_\pi Y \to E\widehat{\otimes}_\varepsilon Y$ is injective when E has the approximation property.

This is not a full solution of Question `Gonzalo`. It leaves open the case where neither factor sequence has a norm-null subsequence nor a complemented subsequence whose closed span has the approximation property. In that remaining case, any counterexample must still converge weakly to a nonzero element of the kernel of the canonical map $X\widehat{\otimes}_\pi Y \to X\widehat{\otimes}_\varepsilon Y$.

Relation to the source paper

The source paper proves a partial affirmative answer under hypotheses that force a non-weakly-null tensor sequence to contain an ℓ_1-subsequence, including an unconditional complemented-subsequence hypothesis on one factor. The present argument is different: because the source question assumes weak convergence, it suffices to project onto a complemented AP span of one subsequence and apply the elementary-functionals/injective-kernel argument there. No unconditionality of that subsequence is used.

Novelty check

The cheap run indexes were searched for arXiv:2305.06089, Question `Gonzalo`, complemented basic subsequences, projective tensor products, and weakly null tensor sequences. The existing run packet '2305.06089_gonzalo_ap_subcase' records the global AP/injective-kernel subcase. The source paper records unconditional-complemented and R_p-type subcases. This packet should therefore be reviewed as a small sequence-local strengthening, not as a full answer.

References

- J. Rodriguez and A. Rueda Zoca, *Weak precompactness in projective tensor products*. arXiv:2305.06089.
- R. A. Ryan, *Introduction to Tensor Products of Banach Spaces*, Springer, 2002. Used for standard projective/injective tensor-product facts and the approximation-property injectivity criterion.

Human-review recommendation

Review as a likely valid partial result. The main point to check is the projection step: the weak limit of a subsequence fixed by $(iP) \otimes \mathrm{id}_Y$ is itself fixed by that projection, while its coordinate in $E\widehat{\otimes}_\pi Y$ is killed by the approximation-property argument.